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PRESENTATION

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

(foreign language) Good afternoon, everyone, and welcome to TSMC's Fourth Quarter 2023 Earnings Conference and Conference Call. It's great to see everyone in person once again. This is Jeff Su, TSMC's Director of Investor Relations and your host for today. Today's event is being webcast live through TSMC's website at www.tsmc.com, where you can also download the earnings release materials. (Operator Instructions)

The format for today's event will be as follows: First, TSMC's Vice President and CFO, Mr. Wendell Huang, will summarize our operations in the fourth quarter 2023 and full year of 2023, followed by our guidance for the first quarter 2024. Afterwards Mr. Huang, TSMC's CEO, Dr. C. C. Wei; and TSMC's Chairman, Dr. Mark Liu, will jointly provide the company's key messages. Then TSMC's Chairman, Dr. Mark Liu, will host the Q&A session where all 3 of our executives will take your questions.

As usual, I'd like to remind everybody that today's discussions may contain forward-looking statements that are subject to significant risks and uncertainties, which could cause actual results to differ materially from those contained in the forward-looking statements. Please refer to the safe harbor notice that appears on our press release.

And now I would like to turn the microphone over to TSMC's CFO, Mr. Wendell Huang, for the summary of operations and the current quarter guidance.

Wendell Huang - *Taiwan Semiconductor Manufacturing Company Limited - VP & CFO*

Thank you, Jeff. Happy New Year, everyone. Thank you for joining us today. My presentation will start with financial highlights for the fourth quarter and recap of full year 2023. After that, I will provide the guidance for the first quarter 2024.

Fourth quarter revenue increased 14.4% sequentially in NT dollar or 13.6% in U.S. dollars, as our fourth quarter business was supported by the continued strong ramp of our industry-leading 3-nanometer technology. Gross margin decreased 1.3 percentage points sequentially to 53%, primarily due to margin dilution from 3-nanometer ramp.

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Operating margin decreased 0.1 percentage point sequentially to 41.6%, slightly ahead of our guidance, mainly due to operating leverage on higher revenue. Overall, our fourth quarter EPS was TWD 9.21 and ROE was 28.1%. Now let me move on to revenue by technology.

The 3-nanometer process technology contributed to 15% of wafer revenue in the fourth quarter, while 5-nanometer and 7-nanometer accounted for 35% and 17%, respectively. Advanced technologies, defined as 7-nanometer and below, accounted for 67% of wafer revenue.

On a full year basis, 3-nanometer revenue contribution came in at 6% of 2023 wafer revenue, 5-nanometer was 33% and 7-nanometer was 19%. Advanced technologies accounted for 58% of total wafer revenue, up from 53% in 2022.

Moving on to revenue contribution by platform. HPC increased 17% quarter-over-quarter to account for 43% of our fourth quarter revenue. Smartphone increased 27% to account for 43%. IoT decreased 29% to account for 5%. Automotive increased 13% to account for 5%. And DCE decreased 35% to account for 2%. On a full year basis, smartphone, IoT, DCE decreased 8%, 17% and 16%, respectively. HPC remained flat, while automotive increased 15% in 2023. Overall, HPC accounted for 43% of our 2023 revenue; smartphone, 38%; IoT, 8%; and automotive, 6%.

Moving on to the balance sheet. We ended the fourth quarter with cash and marketable securities of TWD 1.7 trillion or USD 55 billion. On the liability side, current liabilities decreased by TWD 56 billion, mainly due to the decrease in accounts payable. On financial ratios, accounts receivable days decreased 4 days to 31 days, while days of inventory also declined 11 days to 85 days, primarily due to higher 3-nanometer wafer shipment.

Regarding cash flow and CapEx. During the fourth quarter, we generated about TWD 395 billion in cash from operations, spent TWD 170 billion in CapEx and distributed TWD 78 billion for the first quarter '23 cash dividend. Overall, our cash balance increased TWD 154 billion to TWD 1.47 trillion at the end of the quarter. In U.S. dollar terms, our fourth quarter capital expenditures totaled TWD 5.24 billion.

Now let's look at the recap of our performance in 2023. 2023 was a challenging year for the global semiconductor industry, but our technology leadership enabled TSMC to outperform the foundry industry. Our revenue decreased 8.7% in U.S. dollar terms to USD 69 billion or decreased 4.5% in NT terms to TWD 2.16 trillion. Gross margin decreased 5.2 percentage points to 54.4%, mainly reflecting lower overall capacity utilization and 3-nanometer ramp, partially offset by a more favorable foreign exchange rate.

To extend our technology leadership, we continued to expand our R&D investment in 3-nanometer and 2-nanometer development despite a lower revenue base in 2023. Thus, operating margin decreased 6.9 percentage points to 42.6%. Overall, full year EPS declined 17.5% to TWD 32.34 and ROE was 26.2%. On cash flow, we spent USD 30.45 billion or TWD 950 billion in CapEx, while generating TWD 1.7 trillion in operating cash flow and TWD 292 billion in free cash flow. We also paid TWD 292 billion in cash dividends in 2023.

I have finished my financial summary. Now let's turn to our current quarter guidance. We expect our business in the first quarter to be impacted by smartphone seasonality, partially offset by continued HPC-related demand. Based on the current business outlook, we expect our first quarter revenue to be between USD 18 billion and USD 18.8 billion, which represents a 6.2% sequential decline at the midpoint. Based on the exchange rate assumption of USD 1 to TWD 31.1, gross margin is expected to be between 52% and 54%, operating margin between 40% and 42%. This concludes my financial presentation.

Now let me turn to our key messages. I will start by making some comments on our fourth quarter '23 and first quarter '24 profitability. Compared to third quarter, our fourth quarter gross margin decreased by 130 basis points sequentially to 53%, primarily due to the margin dilution from the continued ramp-up of our 3-nanometer technology. We have just guided our first quarter gross margin to be flat sequentially at 53% at the midpoint, primarily as a less favorable foreign exchange rate assumption is offset by product mix changes due to smartphone seasonality.

Looking at full year 2024, given the 6 factors that determine our profitability, there are a few puts and takes I would like to share. On the plus side, we expect our utilization rate to rise in 2024 as our business recovers. However, as we move -- as we have said before, N3 is expected to dilute our gross margin by about 3 to 4 percentage points for the full year of 2024 as the revenue contribution will be much higher than in 2023.

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In addition, we have a strategy so that some of our N3 capacity can be supported by N5 tools given the strong multiyear demand. Such a plan will enable higher capital efficiency in the mid to long term, but requires cost and effort in the near term. Most of this conversion will occur in second half of 2024, and we expect it to dilute our gross margin by about 1 to 2 percentage points in second half of 2024.

Finally, we have no control over the foreign exchange rate, but that might be another factor in 2024. Long term, excluding the impact of foreign exchange rate and considering our global manufacturing footprint expansion plans, we continue to forecast a long-term gross margin of 53% and higher is achievable.

Next, let me talk about our 2024 capital budget and depreciation. Every year, our CapEx is spent in anticipation of the growth that will follow in future years. In 2023, we spent USD 30.4 billion, lower than our prior guidance of approximately USD 32 billion, as we continue to tighten up our capital spending where appropriate, given the near-term uncertainties.

In 2024, our capital budget is expected to be between USD 28 billion and USD 32 billion as we continue to invest to support customers' growth. Out of the USD 28 billion to USD 32 billion CapEx for 2024, between 70% and 80% of the capital budget will be allocated for the advanced process technologies, about 10% to 20% will be spent for specialty technologies and about 10% will be spent for advanced packaging, testing, mask making and others. Our depreciation expense is expected to increase close to 30% year-over-year in 2024, mainly as we ramp up our 3-nanometer technologies.

Finally, let me make some comments on our long-term CapEx and cash dividend distribution policy. At TSMC, a higher level of capital expenditures is always correlated with higher growth opportunities in the following years. In the past few years, we have sharply increased our CapEx spending in preparation to capture or harvest the growth opportunities from HPC, AI and 5G megatrends.

Despite a challenging 2023, our revenue remains well on track to grow between 15% and 20% CAGR over the next several years in U.S. dollar terms, which is the target we communicated back in January 2022 Investor Conference. With our 2024 CapEx guidance of USD 28 billion to USD 32 billion, the rate of increase of our capital spending has begun to level off as we capture and harvest the growth. The objectives of TSMC's capital management are to fund the company's growth organically, generate good profitability, preserve financial flexibility and distribute a sustainable and steadily increasing cash dividend to shareholders.

As a result of our rigorous capital management, in November, TSMC's Board of Directors approved the distribution of a TWD 3.5 per share cash dividend for the third quarter of 2023, up from TWD 3 previously. This will become the new minimum quarterly dividend level going forward. Third quarter '23 cash dividend will be distributed in April 2024.

In 2023, TSMC's shareholders received a total of TWD 11.25 cash dividend per share, and they will receive at least TWD 13.5 per share cash dividend for 2024. In the next few years, we expect the focus of our cash dividend policy to continue to shift from a sustainable to a steadily increasing cash dividend per share. Now let me turn the microphone over to C. C.

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

Thank you, Wendell. Good afternoon, everyone. First, let me start with our 2024 outlook. 2023 was a challenging year for the global semiconductor industry, but we also witnessed the rising emergence of generative AI-related applications with TSMC as a key enabler.

In 2023, weakening global macroeconomic conditions and high inflation and interest rate exacerbated and prolonged the global semiconductor inventory adjustment cycle. Concluding 2023, the semiconductor industry excluding memory industry declined about 2%, while foundry industry declined about 13% year-over-year. TSMC's revenue declined 8.7% year-over-year in U.S. dollar terms.

Despite the near-term challenges, our technology leadership enable TSMC to outperform the foundry industry in 2023, while we are positioning us to capture the future AI and high-performance computing-related growth opportunities.

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Entering 2024, we forecast fabless semiconductor inventory to have returned to a healthier level exiting 2023. However, macroeconomic weakness and geopolitical uncertainties persist, potentially further weighing on consumer sentiment and the market demand. Having said that, our business has bottomed out on a year-over-year basis, and we expect 2024 to be a healthy growth year for TSMC, supported by continued strong ramp of our industry-leading 3-nanometer technologies, strong demand for the 5-nanometer technologies and robust AI-related demand.

Coming off the steep inventory correction and low base of 2023, for the full year of 2024, we forecast the overall semiconductor market excluding memory to increase by more than 10% year-over-year, while foundry industry growth is forecast to be approximately 20%. For TSMC, supported by our technology leadership and broader customer base, we are confident to outperform the foundry industry growth. We expect our business to grow quarter-over-quarter throughout 2024, and our full year revenue is expected to increase by low to mid-20% in U.S. dollar terms.

Next, let me talk about our N3 and N3E ramp-up and progress. Our 3-nanometer technology are the most advanced semiconductor technology in both PPA and transistor technology. As a result, almost all the world's smartphone and HPC innovators are working with TSMC on 3-nanometer technologies. Our N3 successfully entered volume production and enjoy a strong ramp in second half '23, accounting for 6% of our total wafer revenue in 2023.

N3E further leverage the strong foundation of N3 to extend our N3 family with enhanced performance, power and yield. N3E has already entered volume production in the fourth quarter of 2023. Supported by robust demand from customers in both smartphone and HPC applications, we expect revenue from our 3-nanometer technology to more than tripled in 2024 and account for a mid-teens percentage of our total wafer revenue.

We also continue to provide further enhancement of our N3 technology, including N3P and the N3X. With our strategy of continuous enhancements of our 3-nanometer process technologies, we expect strong multiyear demand from our customers and are confident that our 3-nanometer family will be another large and long-lasting node for TSMC.

Now I will talk about the AI-related demand and our N2 status. The surge in AI-related demand in 2023 supports our already strong conviction that the structural demand for energy-efficient computing will accelerate in an intelligent and connected world. TSMC is a key enabler of AI applications. No matter which approach is taken, AI technology is evolving to use more complex AI models as the amount of computation required for training and inference is increasing.

As a result, AI models need to be supported by more powerful semiconductor hardware, which requires use of the most advanced semiconductor process technologies. Thus, the value of TSMC technology position is increasing, and we are all well positioned to capture the major portion of the market in terms of semiconductor component in AI. To address insatiable AI-related demand for energy-efficient computing power, customers rely on TSMC to provide the most leading edge process technology at scale with a dependable and predictable cadence of technology offering.

At the same time, as process technology complexity increase, the engagement lead time with customers also started much earlier. Thus, almost all the AI innovators are working with TSMC, and we are observing a much higher level of customer interest and engagement at N2 as compared with N3 at a similar stage from both HPC and the smartphone applications.

Our 2-nanometer technology will adopt ~~narrow sheet~~ nanosheet transistor structure and be the most advanced semiconductor technology in the industry in both density and energy efficiency when it is introduced in 2025. Our N2 technology development is progressing well with device performance and yield on track or ahead of plan. N2 is on track for volume production in 2025 with a ramp profile similar to N3. As part of our N2 technology platform, we also developed the N2 with backside power rail solution, which is better suited for specific HPC applications based on performance, cost and maturity considerations.

N2 with backside power rail will be available in the second half of 2025 to customers with production in 2026. With our technology of continuous enhancement, N2 and its derivative will further extend our technology leadership position and enable TSMC to capture the AI-related growth opportunities well into the future.

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Finally, let me talk about our specialty technology strategies at mature node. For TSMC, today, around 70% of our total revenue is 16-nanometer and more advanced node. With rising contribution from 3-nanometer and 2-nanometer technologies in the next several years, this number will only increase. Thus, our mature node exposure is around 20% of our total revenue.

TSMC's strategy at mature node is to work closely with strategic partner to develop specialty technology solutions to meet customers' requirement and create differentiated and long-lasting value to customers. Our focus is to build a high yield capacity for specialty technologies rather than just a nominal capacity. Through the development, deployment of the differentiated specialty technologies, the profitability of our mature nodes can be around our corporate average gross margin.

Looking ahead, we forecast 28-nanometer will be the sweet spot for our embedded memory applications, and we expect our long-term structural demand at 28-nanometer to be supported by multiple types of specialty technologies. Thus, we are expanding our 28-nanometer specialty manufacturing capacity overseas to support the long-term structural market demand. We believe demand for the differentiated specialty technology will remain steady despite the potential industry capacity increase and our utilization rate and structural profitability at mature node can be well protected in the future.

This concludes my prepared remarks, and now let me turn the microphone over to Mark.

Mark Liu - *Taiwan Semiconductor Manufacturing Company Limited - Chairman*

Thank you, C. C. Good afternoon, everyone. First, let me talk about our global manufacturing footprint update. TSMC's mission is to be the trusted technology and capacity provider for the global logic IC industry for years to come.

In today's fractured globalization environment, our strategy is to expand our global manufacturing footprint to increase our customer trust, expand our future growth potential and reach for more global talents. Our overseas decisions are based on our customers' needs and the necessary level of government subsidy or support. This is to maximize the value for our shareholders.

Firstly, in Japan, we are building a specialty technology fab in Kumamoto, which will utilize 12- and 16-nanometer and 22- and 28-nanometer process technologies. We will hold an opening ceremony for this fab on February 24, next month, and volume production is on track for the fourth quarter of 2024.

In Arizona, we are in close and constant communication with the U.S. government on incentive and a tax credit support and making strong progress in facility supply chain infrastructure, utility supply and equipment installation for our first fab. We continue to work closely and develop strong relationships with our local union and trade partners in Arizona, including recently signed an agreement with Arizona Building and Construction Trades Council on a new framework for cooperation. This agreement extends our collaboration across enhanced workforce training and development, shared commitment to site safety, hiring local workers and establishing regular communication. It is a win-win for all parties.

We are well on track for volume production of N4 or 4-nanometer process technology in first half of '25 and are confident that once we begin operations, we will be able to deliver the same level of manufacturing quality and reliability in Arizona as from our fabs in Taiwan.

In Europe, we plan to build a specialty technology fab in Dresden, Germany, focusing on automotive and industrial applications with our joint venture partners. We continue to be in close communication with the German federal, state and city governments and their commitment to this project remain strong and unchanged. Fab construction is scheduled to begin in Q4 2024 this year.

In Taiwan, of course, we continue to invest in and expand our advanced technology capacities to support our customers' needs and their growth. Given the robust multiyear demand for our 3-nanometer technologies, we are expanding our 3-nanometer capacity in Tainan Science Park. We are also preparing our N2 volume production starting in 2025. We plan to build multiple fabs or multiple phases of 2-nanometer technologies in both Hsinchu and Kaohsiung science parks to support the strong structural demand from our customer C. C. just mentioned. In Taichung Science Park, the government approval process is ongoing and is also on track.

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While the initial cost of overseas fab, I previously mentioned, are higher than TSMC's fab in Taiwan, we are confident to manage and minimize the cost gap and remain committed to deliver profitable growth and maximize the value for our shareholders.

Now let me talk about my retirement. On December 19 last year, I announced that I have decided not to seek nomination of Board members for the next term and will retire from the company after the 2024 Annual Shareholders Meeting in June. Allow me to say this. Over the past 30 years, I have been incredibly fortunate to be able to work at and contribute to TSMC.

I started at TSMC 30 years ago as a leader of a small 4-person fab construction team. It has been my privilege to serve as Chairman of TSMC and after our legendary founder, Dr. Morris Chang, over the last 6 years. During this time, we have reaffirmed our commitment to our mission, to be the trusted technology and capacity provider to the global logic IC industry for years to come, while adhering to our core values of integrity, commitment, innovation and customer trust.

TSMC's success is predicated on providing the industry's most leading-edge process technology at scale in the most efficient and cost-effective manner, to enable all the innovators to successfully offer their best products to the world. We, together, have worked diligently to enhance our focus on our technology leadership, competitiveness, global manufacturing footprint, digital excellence, sustainability, and corporate governance to maximize the value for our customers and our shareholders.

The past 30 years with TSMC has been an extraordinary journey for me, and I want to extend my sincerest thanks to our incredibly talented team and all our TSMC's colleagues, whose diligence, dedication and "can-do" spirit have made the company into what it is today.

Now TSMC's Nomination, Corporate Governance and Sustainability Committee of the Board has recommended Dr. C. C. Wei to succeed as the company's next Chairman, subject to the election of the incoming Board in June 2024. If Dr. Wei is elected to be Chairman, he should also continue in his current role as CEO, supported by a deep and experienced team of senior executives, many of whom have been with TSMC for many, many years.

As I look ahead to spend more time with my family and starting the next chapter of my life after our AGM in June, I remain fully confident in TSMC's strategy, leadership and execution, and firmly believe TSMC will continue to perform outstandingly in the years ahead. Thank you for your trust in TSMC and the best is yet to come for the company and its shareholders.

This conclude my messages and our key messages together. Thank you for your attention.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Thank you, Chairman. This concludes our prepared statements.

QUESTIONS AND ANSWERS

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

(Operator Instructions) So now let's begin the Q&A session. Again, our Chairman, Dr. Mark Liu, will be the host. Let's take the first 2 questions from the floor, please. Okay. Our first question comes from Charlie Chan from Morgan Stanley.

Charlie Chan - *Morgan Stanley, Research Division - Technology Analyst*

It's great to see you again in person, Happy New Year. Allow me to remain seated. I have a some long question to you. So first question is to C. C. I am very curious about your comment about the technology leadership, right, because your competitor and also customer, Intel, states that their PPA is ahead of your 2-nanometer, even the cost is lower.

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So I want to consult your opinion why there's a different story and how do we judge? And given this debate, how TSMC is going to plan the future capacity for this customer and also competitor? We want to seize this opportunity, but also avoid any over-expansion.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Thank you, Charlie. Just please allow me for the benefit of the audience here in person and online to summarize your question. So Charlie's first question is around sort of the technology leadership and also our relationship, I guess, our capacity planning with a specific IDM.

So the first part of his question is on the technology part. He notes this IDM says their PPA is ahead of TSMC's 2-nanometer and the costs can be lower, yet we said our technology is industry leading, so how do we reconcile the difference? And also, how do we plan the future capacity planning for such type of customer?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

Charlie, you named my customer's name. That's my customer and my competitor. Let me repeat the last time when I comment on their technology. The comment stays the same, so that their newest technology will be very similar or equivalent to TSMC's N3P. We further checked again with all the specs, all the possible published in the technology, transistor technology and everything. My comment stays the same.

With a big advantage in the technology maturity, because of in 2025, when they say that their newest technology will go on production, for TSMC, that will be the third year with a very high volume production in the fabs. So again, I don't want to make too much of a comment on my customers' claim. But let me assure you, we continue to have a technology leadership and we continue to have a broad base of customers. And almost everybody, almost, they are working with TSMC, okay?

Charlie Chan - *Morgan Stanley, Research Division - Technology Analyst*

In that case, would you aggressively spend capacity because outsourcing is more likely?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

Certainly, we expand our capacity with USD 28 billion to USD 32 billion, that's a big money. That will be used for 3 nanometer and 2 nanometer's capacity.

Mark Liu - *Taiwan Semiconductor Manufacturing Company Limited - Chairman*

Let me add some color to this, C. C. I think C. C. has been very modest. I think he's claimed that our N3P is comparable to their 18A. We still affirm our statement. But I would like you to look at a different perspective. And what C. C. -- what the other side claim might be right, but it's only to their own product. An IDM typically optimize their technology for their own product, where foundry, us, we optimize our technology for our customers' products, so that's a big difference.

What you use for the high-power server could be very different than what is used with these gadgets on your hands, smartphone or even the large edge AI processors, so you should look at this. I think the time -- compared with PPA, we still affirm our statement. But I think just look at our customers' action, that just tell this -- all the stories.

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Charlie Chan - *Morgan Stanley, Research Division - Technology Analyst*

So Jeff, can I go for a second one? Yes. So Mark, so first of all, I really appreciate your leadership. I believe the global investors appreciate your past 6 years, create lots of shareholders' value. So my question is about the content of your speech in November. The speech was about the TSMC in the era for AI. And you mentioned some very interesting data points. You use the AI technology to improve the defense [cloudification].

Also, the EUV throughput by, for example, 10%, right? So now it's -- the generative AI can be very big breakthrough in terms of technology. Do you think Samsung or Intel, by leveraging the generative AI, can really break through and catch up your technology? And also before your retirement, any kind of a big unfinished goal or targets for TSMC?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Thank you, Charlie. So Charlie's second question is directed to Chairman. He noted in November, Chairman gave a speech where he shared how TSMC has always been utilizing big data, machine learning and AI to improve our operational efficiencies. His question is whether now with generative AI, will this enable or allow our competitors to do the same thing and catch up and narrow the gap. Yes.

Mark Liu - *Taiwan Semiconductor Manufacturing Company Limited - Chairman*

Okay. Thank you, Charlie. The talk I gave in last November was -- the audience is the Taiwan's industry, companies. The purpose I want to give that is I see artificial intelligence can be a great opportunity for the industry in Taiwan. Just like Taiwan is a big country for semiconductor, it can be a big country for artificial intelligence in the future. That's how I encouraged them.

And as far as the -- whether our competitor are using AI, of course, of course, they use AI. And just look at all the company -- AI company in Silicon Valley or in U.S. That's not a secret. But on the other hand, AI is only in its nascent stage. Only last November, the first large language data is announced, ChatGPT announced.

We only see the tip of the iceberg. So I want to give the industry an optimistic note that even though 1 nanometer or sub 1 nanometer could be challenging, but we have a new technology capability using AI to accelerate the innovation in science. And that is the -- that is our part, and we have been working on that for many years already. So we opened -- of course, it's a fair competition. It's no secret.

You have another question about one. Well, I still have -- yes, indeed, I will retire in June. And from now to June is a long time. And the company, a lot of things can happened. And I think -- I hope we definitely execute to C. C.'s forecast of this year. I think by the middle of this year, I think we're pretty sure we can accomplish that.

And of course, C.C. just mentioned, our technology development is on the slew of success. And by June this year, we will know what we are going to fare in 2025. And I give our executive a milestone. I don't want share with you, but it's going to be very exciting for TSMC.

And of course, from now on, I simply want to encourage our people in TSMC that the world has changed. Just like you mentioned, we have to use artificial intelligence for future technology. So we will go into global -- we try our global footprint, and we are trying our digital excellence. By digital excellence, it means we can't count on the hardworking of Taiwan engineers only. We have to recreate our job to tap their talents and lift up the semiconductor technology engineering to a different level based on what we already have.

And of course, the corporate governance is one thing I always see in my heart. During this transition, I want every executive and our Board to adhere to the sound corporate governance so that make sure all the process steps is abided by our ethical governance rules. Thank you.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Thank you, Chairman. Let's take the -- move on to the second person from the floor. I think in front, Bruce Lu of Goldman Sachs, please.

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Bruce Lu - *Goldman Sachs Group, Inc., Research Division - VP*

Again, the question is definitely coming from the AI, for sure. I think, as C. C. mentioned, almost every AI chip is working with TSMC. However, the investors' concern is always like the dollar content, as a percentage of customers' cost for AI is a lot lower than smartphone or other chips. So as also C. C. mentioned, that you sell a wafer for your customer, but when you buy it back, it's a lot more expensive.

So can we expect that the dollar generated by TSMC from AI can be increased in the coming years, whether it's through, like, node migration or advanced packaging or anything we can expect or what kind of rate we can expect for that?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Let me summarize Bruce's first question. I think, again, he is around AI related. He notes that basically, almost all the innovators are working with TSMC at AI, but the value per chip that we seem to be capturing is lower than for a smartphone or PC. So his question is that I think can we expect the dollar value captured by TSMC to increase in the next few years? And will this be -- this additional value more come from the front-end process node wafer production? Or will it be through the advanced packaging solutions?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

Well, let me answer the easiest one first. The revenue come from the front-end and back end together. Okay. To capture the value, yes, we are working on it, de-initely. But first, let me say that I'm very happy that my customer has been very successful in the AI area, and we are a key enabler for the AI applications.

So far today, everything you saw on the AI came from TSMC, okay? Now here comes the question is that how are we going to capture the value. We are working on it. And actually, we see today, let's say that the component total value in the the whole AI data center is a very small percentage amount. If we narrow down the AI's component or the semiconductors' value in the whole system, yes, it's a small percentage.

But for TSMC, we look at ourself, the AI's a CAGR, that's the growth rate every year, it's about 50%. And we are confident that we can capture more opportunities in the future. So that's what we said that up to 2027, we are going to have high teens of the revenue from a very narrowly defined the AI application processor, not to mention about the networking, not to mention about all others, okay?

And to further extend our value, actually, all the edge device, including smartphone, including the PC, they start to put the AI's application inside. They have some kind of a neural processor, for example, so the silicon content will be greatly increased. Although the unit is actually low single digit in CAGR, but the silicon content is more important. So put all together, if we lumpsum all the AI-related application, actually, it's quite a big amount for TSMC to grow.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Bruce, do you have a second question, please.

Bruce Lu - *Goldman Sachs Group, Inc., Research Division - VP*

Yes. The second question is more for the technology leadership. I mean, as we move into the nanosheet or advanced node, we see another technology divide nowadays. For example, like a high-NA EUV tools. TSMC seems has a different view with other peers. I mean, in the past, like, 20, 30 years, there are several technology divide that TSMC always choose the right decisions, right? So can you tell us about why you choose your current route comparing to your peers? What is the pros and cons? And what's the advantage? And how confident that TSMC can leverage that to be the key success factor for the leading edge?

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Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Bruce's second question is in regards, I think, to technology development and decision-making basically. He notices that today, there's divergence or his words, divide between different companies technology decision, whether to adopt nanosheet transistor structure, whether to adopt high NA tools. He notes that in the past, that this is always occurred in our industry, but TSMC has somehow managed to make the right decision.

So he is asking, especially how -- what do we look at or evaluate in our decision-making process, what are the pros and cons and advantages, and probably most importantly, how confident are we about our technology decisions going forward, whether nanosheet, high NA given our competitors' actions?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

Bruce, you asked a very technical question. I'm not very sure everybody knows the high NA or is the nanosheet or gate all around. But let me answer the question. We always make the right decision, and our track record shows that. Is that enough?

Okay. Let me elaborate a little bit more. Because of technology itself is no value. Only when that can serve your customer. So we always work with our customers to give them the best transistor technology and the best power-efficient technology and at a reasonable cost, okay? And more importantly, the technology maturity that -- in the high-volume production, that's all important.

Everything. Everything counted together. So we -- every time we know that there are some new structure, new tools such as high-NA EUV, we look at it carefully, look at the maturity of the tools, look at the cost of the tools and look at the schedule of that -- how to achieve it. We always make the right decision at the right moment to serve our customers.

And so far, all our customers are happy with TSMC's progress, okay? Did that answer your question. Almost everybody worked with TSMC on 2-nanometer, except 1.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Thank you, C. C. All right. Let's go to the online, take the next 2 questions from the participants who are dialing in via the conference call, please. Operator, could you please state the name and company?

Operator

The first person to ask a question from the line is Gokul Hariharan from JPMorgan.

Gokul Hariharan - *JPMorgan Chase & Co, Research Division - MD, Co-Head of Asia TMT Research, Head of Taiwan Equity Research & Senior Tech Analyst*

Yes. (inaudible)

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Gokul, okay, I need you to slow down a little bit because the line is not that clear. I do think I got his question, which is he wants to confirm, C. C., you mentioned that we have a very narrow definition, we call server AI processor contribution and that you said it can be high teens in 5 years' time because the last time, we said low teens.

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C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

The demand suddenly being increased since last -- I think, last year, the first quarter up to March or April, when ChatGPT become popular, so customers respond quickly and asked TSMC to prepare the capacity, both in front-end and the back-end. And that's why we have confidence that this AI's revenue will increase.

We only narrowed down to the AI application processor, by the way. So we look at ours here, that we prepare the technology and the capacities in both our front-end and also back-end. And so we -- it's in the early stage so far today. We already see the increase, the momentum. And we expect -- if you guys continue to track this one, the number will increase. I have confidence to say that, although I don't know how much.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

So high teens, you confirm?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

Or higher.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Yes. Okay. So okay, Gokul, hopefully, that clarifies that first question.

Gokul Hariharan - *JPMorgan Chase & Co, Research Division - MD, Co-Head of Asia TMT Research, Head of Taiwan Equity Research & Senior Tech Analyst*

The second is about gross margin. In the downturn, we are forming out at much higher gross margin levels than before. Talk a little bit about what happens when we get back to close to full utilization. So I think we are still running at well below full utilization in 2023. And could you also explain the gross margin dilution that you're expecting in second half '23 because of this capacity conversion? What exactly leads to that gross margin dilution? And is that like a onetime kind of dilution that lasts for a little bit of time and kind of levels off in 2025?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Thank you, Gokul. So if I heard correctly, Gokul's second question is around gross margin. So 2 parts to it. Maybe the second part first, which is he is asking, I believe, about this gross margin in the second half of this year, particularly with what Wendell had described. There are plans to convert some of the capacity, the gross margin impact here, and is this a onetime thing? Is this better capital -- what does this mean in the mid- to long-term profitability? That's the first part. And then I'll go to the second one.

Wendell Huang - *Taiwan Semiconductor Manufacturing Company Limited - VP & CFO*

Right. Second half, the -- as I said, there are 2 negative factors affecting our gross margin this year. The first one is the N3 dilution. N3 volume will be much bigger in the second half than the first half. So the second half impact from N3 dilution will be between 3 to 4 percentage points.

And also the N5 capacity converted to N3. That will mostly take place in the second half as well. So that will be 1 to 2 percentage points, okay? That's for this year. For the longer term, if you look at these 2 factors, our N3 dilution will gradually reduce, because the profitability will continue to improve or increase in the next several years.

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And N5 converted to N3, it's a onetime short-term impact on profitability, which will bring capital efficiency to us in the middle to long term. And the benefits together will be much bigger than the onetime hit in the short term. So if you're talking about the longer-term profitability, including these 2 factors, plus we're selling our value, our technology value, as C. C. mentioned, we continue to drive down the cost.

We build our capacity based on the long-term market profile and not the short-term cyclicality and therefore, enable us to have a pretty good utilization. The only thing we are not able to control is foreign exchange rate. So if you put all these together, we still believe that 53% and higher long-term gross margin is achievable.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Gokul, does that answer both parts of your question?

Gokul Hariharan - *JPMorgan Chase & Co, Research Division - MD, Co-Head of Asia TMT Research, Head of Taiwan Equity Research & Senior Tech Analyst*

Yes. So just to clarify, so given we are much -- at a much lower utilization than normal, what you suggests, Wendell, is that gross margin should get back to the mid- to high 50s once the up cycle starts to gain more momentum, just like what we saw in 2022. Is that a reasonable expectation?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Yes. Okay. Thank you, Gokul. So Gokul, really, he's asking 53% and higher. Can it be higher? Because, of course, he looks at last year, the utilization was lower, and we still managed to deliver. So he's wondering once utilization goes back to full, can it get to mid- to high 50s?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

We are working on it. Okay. Certainly, we prepare our capacity according to customers' demand. Last year is very challenging because everybody missed their forecast and so did TSMC. And so the utilization rate is pretty bad. And I believe everybody got more experience in the next few years. And so TSMC's utilization rate will continue to increase, I guarantee that.

Mark Liu - *Taiwan Semiconductor Manufacturing Company Limited - Chairman*

So the question is we're working on that. It can be.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Thank you, Gokul. Thank you. Operator, let's move on to the -- take the question from the second participant on the call.

Operator

The second one to ask a question is Randy Abrams from UBS.

Randy Abrams - *UBS Investment Bank, Research Division - Analyst*

Okay. Yes. And good luck to both Mark and C. C. as you go through the upcoming transition. I wanted to ask, going back to the question on the IDM. I think, earlier, you conceded that your competitor's process is actually pretty good for -- optimized to their own products. Could you talk about your view on sustainability of the ramp of that IDM outsourcing with your own products in HPC? If you look out over the next 2 to 3 years,

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if you see that continuing to grow or reversed, or there could be a bit of a cooling off from some of the opportunity you have right in front of you now?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Thank you, Randy. So Randy's first question goes back to the IDM. His question is, with the IDM saying their technology is pretty good, what is the risk? Or how do we see the sustainability of this IDM's outsourcing business to TSMC in the next 2 to 3 years? Can this continue to grow? Or will this reverse and go back in-house to the IDM? And how do we manage or plan for this?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

Okay. Randy, that's a good question. Actually, we have taken into account all the considerations, including the IDM, can do it by themselves. We have put that one into consideration, actually, in our capacity planning, actually, we took a very conservative way to prepare our capacity in this kind of a situation. Okay. I cannot speak more because of that's our strategy.

Randy Abrams - *UBS Investment Bank, Research Division - Analyst*

Okay. If I can ask a follow-up, actually, just through the CapEx, where I think, earlier, you said the rate of increase would slow down. But I think still implying it should increase over time as you grow. If you could discuss -- the CapEx that you guided was flat. Should we think of it as a pause whereas you start to move into 2-nanometer, there should be another wave of increase?

And second part, somewhat related, but curious about the geographic expansion. There's been a lot of press about new fabs in Japan, second fab and potential third advanced fabs. And it feels like the first fab went smoothly. So are you starting to redirect or think more expansion to Japan rather than U.S.? Or potentially both, you have both options as you move to 3-nanometer?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay, Randy. That's a lot of questions. So I'm going to take that as your second question, okay, basically. So the first part of it is about the CapEx. He notes that Wendell said the rate of increase is beginning to level off. Randy's question is for this year and take the midpoint, $30 billion is basically flat. So is this just a temporary pause in the CapEx? And with 2-nanometer in the upcoming years, should we expect the dollar amount to go back up? That's the first part.

Wendell Huang - *Taiwan Semiconductor Manufacturing Company Limited - VP & CFO*

Okay, Randy, the CapEx dollar amount, every year, may vary. It depends on the different situation. The rate of increase definitely is slower than the past 3 years. And if you look at -- I think the other way of looking at that is the capital intensity. In the past 3 years, the highest point is 2021. So it's going to be -- it was over 50% and then followed by 47% and 43%. And this year, if you do the math, is going to be mid-30s. We expect, in the next several years, it will remain around mid-30 percentage capital intensity.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. And then the other part of Randy's question is on the geographic expansion. He notes a lot of reports saying we may build a second fab in Japan and that we even may build a third. So his question is really, are we redirecting our overseas expansion focus more to Japan? Or has it changed anything in the U.S.? Randy, I think that's what you're trying to ask, right?

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Randy Abrams - *UBS Investment Bank, Research Division - Analyst*

Yes, that's what I'm trying to ask.

Mark Liu - *Taiwan Semiconductor Manufacturing Company Limited - Chairman*

Can you repeat the question again?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Sure. So Randy is saying, look, he knows there's a lot of talk, we're going to build a second fab in Japan, maybe 3. So he just wants to know, are we shifting our overseas expansion focus to Japan or -- from the U.S.? Or is there any big significant change?

Mark Liu - *Taiwan Semiconductor Manufacturing Company Limited - Chairman*

No, no. I think you -- Japan, the second fab in Japan is in serious evaluation stage. We haven't announced to the public yet. And we're still discussing with the Japan government. Although many -- they are very cooperative, so we might be waiting for that. But that technology will still be either 7 or 16/12 technologies. And remember the -- our Kaohsiung fab, the first fab used to be 28-nanometer or 7-nanometer. Now it's becoming 2-nanometer. So that is the shift to -- for the -- if there is a second fab in Japan, that's our current plan. And as far as the...

Randy Abrams - *UBS Investment Bank, Research Division - Analyst*

That's helpful. And for -- and to quickly talk 3-nanometer because 5-nanometer was slightly delayed, would 3-nanometer still come 2 years after the new plan for 5 in Arizona?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

So in Arizona, Randy wants to know that we have 5-nanometer in first half '25. What's the plan for the second fab with 3-nanometer?

Mark Liu - *Taiwan Semiconductor Manufacturing Company Limited - Chairman*

Yes, we are -- the second fab shell is under construction. But what technology in that shell is still under discussion, and I think that also has to do with how much incentives that fab -- the U.S. government can provide. And yes, there will be a gap, at least current planning is '27 or '28, yes. That will be the timeframe.

To be honest, mostly -- most all the fab in overseas, what's actually being loaded, what technology being set up, really, it's a decision of customers' demand in that area at that timing, so nothing is definitive. But we are trying to optimize the value for the overseas fab for TSMC.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Thank you, Chairman. Thank you, Randy. All right. In the interest of time, we'll take the next 2 from the floor. I think there's one here. First, Laura Chen from Citi Bank.

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Laura Chen - *Citigroup Inc., Research Division - Research Analyst*

I think we got a lot of discussion about the leading position in the most advanced nodes. So I just have a question about what's your view on the mature node dynamic? And particularly, we are seeing that globally, consider the geopolitical tension. So we are seeing that the fab, all over the place in the world, so do you see that in the longer term, any concern on industry-wise overcapacity? So what's TSMC's strategy? And also, what's your view on your mature node profitability as well? That's my first part.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Thank you, Laura. So Laura's first question is on mature node strategy and profitability. She notes, with the geopolitical dynamics, that there's a lot of capacity being built on the mature nodes. So her question is do we see or expect an industry-wide oversupply, and probably more importantly, what is the impact to TSMC's mature node strategy and profitability?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

Laura, I think your observation is right. There might be too much of a capacity being built right now for mature nodes. So the concern on overcapacity is valid. Now let's talk about TSMC. As I said, the TSMC increased the mature node capacity for specialty technology, differentiated with others. We work with customers and that kind of capacity, actually, effective capacity as we name it, is with a commitment from customers' loading and for the future business.

And because we offer the value for our customer to design their product, so we believe that they can retain their product's value even the capacity is flooded in the industry. And so long as our customer is doing well, TSMC is doing well. And so the profitability, as I said in my statement, it will be around the corporate average, so we don't have concern.

Mark Liu - *Taiwan Semiconductor Manufacturing Company Limited - Chairman*

We speak for TSMC, okay? It could be industry issues.

Laura Chen - *Citigroup Inc., Research Division - Research Analyst*

That's very helpful. And also my second question is back to AI related. As we know that a lot of investors care a lot about your advanced packaging progress. We also know that TSMC got a very good progress on 3D IC SoIC, so can you share with us your progress development beyond CoWoS, what's your plan on the 3D IC and what's the schedule and the capacity you are aiming for in the next 2, 3 years?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Thank you, Laura. So Laura, second question is on advanced packaging. She notes again the strong demand for AI-related applications. So advanced packaging, the progress. Of course, CoWoS, demand is very strong. Her question is, really, I think, beyond CoWoS, into true 3D IC or integration solutions, such as SoIC. What is the progress that we see the engagement from customers, the capacity and basically the outlook for these segments of the business?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

The demand actually is very strong. Today's situation, that we cannot offer enough capacity to support our customers, and that condition will continue probably all the way to next year, although we are very -- working very hard to increase the capacity. For example, this year, we are doubling our output, still not enough. And so we continue to increase for the next year.

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The progress, so far so good because we invested on the advanced packaging technology for more than 10 years, already. So we expect the growth rate for CoWoS for 3D IC or for SoIC per se, it will be more than 50% CAGR in the next few years, at least. And so we are confident that the demand is there. It's TSMC's capability to offer enough capacity to support our customers.

Laura Chen - *Citigroup Inc., Research Division - Research Analyst*

For CoWoS, you will be doubling. And what's the idea about the next year? Do you have any preliminary thought?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

I will talk to you next year.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Thank you. We have a question here from Brad Lin from Bank of America Merrill Lynch. I think in the interest of time, we'll take 1 question. Sorry, from yourself, and then we'll take 1 more from the line and then 1 more in person if there's any.

Brad Lin - *BofA Securities, Research Division - Research Analyst*

Well, so my question will be still around N3 and also IDM. So as we understand, the demand is uncertain, but we can definitely increase our business certainty by gaining market share. So do we expect some more contribution or market share gain, especially from IDM side? Or any more contribution from PC side, maybe, well, by the end of the year or any time soon?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. So Brad's first question is about IDM outsourcing, I think, again, given the technology leadership that we have, he wants to know, do we expect more business or outsourcing from the IDM by the end of this year? Or -- and how do we see it? Is it uncertain going forward?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

That is too specific. You say that IDM outsourcing, I know whom you talk about. So I better not to make any comment. I --stay with what I said, we take everything into consideration. We welcome the business, but we prepared our capacity expansion, okay?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Your second question?

Brad Lin - *BofA Securities, Research Division - Research Analyst*

Yes. So it's on the advanced packaging. So we know that the CoWoS-S, as right now, it's the mainstream. So have the management seen the clients converting to either CoWoS-R or CoWoS-L? And then what's the implication to revenue and margin profile?

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Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. So it's also a very specific, too specific question. But again, Brad wants to know, CoWoS-S seems to be the mainstream today. Do you see customers switching to CoWoS-L or CoWoS-R? And what's the margin implication?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

Well, let's make a joke. I even didn't know what is CoWoS-R or CoWoS-L. But anyway, we are working with customers to support them with adequate capacity, although it's not 100% enough, but we do our best. And we're developing that next-generation CoWoS-L or something like that for our customer. And it's overwhelming that it's welcomed by all my customers, so we are preparing the capacity for it.

Brad Lin - *BofA Securities, Research Division - Research Analyst*

Got it. Last one, not the least, not a question.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

That's 2 questions, so...

Brad Lin - *BofA Securities, Research Division - Research Analyst*

No, no, not a question. So basically, well, I want to say, well, thank you, Mark, for your leadership, contribution, endeavor for the past 30 years, not just for Taiwan -- not just for TSMC, but also for Taiwan. And I wish you a happy retirement and also the new chapter of life.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. We will take the final questions from the last 2 participants. Let's go online first, then we have 1 in final in person, okay? Operator?

Operator

Next from the call is Krish Sankar from TD Cowen.

Krish Sankar - *TD Cowen, Research Division - MD & Senior Research Analyst*

I have 2 of them. First one, I think, Wendell, you spoke about revenue growth for the year and again from gross margin guidance. I'm just kind of wondering how do you think about gross margin for the full year in the context of the fact that TSMC's revenue is going to grow in the low to mid-20%. How do you think about gross margin for full year 2024? And then I have a follow-up.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Krish, again, sorry, we could not hear you that clearly online. But I think his question is, correct me if I'm wrong, with the revenue outlook that we gave, low to mid-20s growth in U.S. dollar term, what is the outlook for the full year gross margin? Is that what you're asking?

Krish Sankar - *TD Cowen, Research Division - MD & Senior Research Analyst*

That's correct.

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Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Yes. Okay.

Krish Sankar - *TD Cowen, Research Division - MD & Senior Research Analyst*

That's right, Jeff. Yes.

Wendell Huang - *Taiwan Semiconductor Manufacturing Company Limited - VP & CFO*

Right. So I just mentioned a couple of puts and takes on the gross margin of this year. And I also said that the second half, we will have a higher dilution from 2 factors. But we're not ready to give out a full year guidance on gross margin yet. So we'll talk about that as time goes by. But let me say this, longer term, with all the factors together, still 53% and higher is definitely, we're very confident in achieve that.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Does that -- yes. Sorry, Krish, why don't you go ahead?

Krish Sankar - *TD Cowen, Research Division - MD & Senior Research Analyst*

I have a follow-up. Yes, just a quick follow-up. And I also just want to say thanks a lot, Mark, for all the support. Just a follow-up. In terms of the revenue growth for this year, December quarter, you exited HPC and smartphone roughly 43% of revenue. What's going to drive the growth this year? Is it HPC? Or smartphone? Which is going to be better this year to get to the low to mid-20%?

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Krish, sorry. Again, we could not hear you that clearly, but I think I got the gist of your question. Maybe the way Krish -- his question is what the components that's driving the revenue growth this year. Maybe we can share with him by the 4 growth platforms.

Wendell Huang - *Taiwan Semiconductor Manufacturing Company Limited - VP & CFO*

Krish, the HPC will be -- will have the highest growth, actually much higher than the corporate. The other 3 platforms will all grow, although slower than the corporate.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Okay. Then we'll take the final question from the floor, the first row here. Nicolas Baratte of Macquarie.

Nicolas Baratte - *Macquarie Research - Analyst*

Very quick question. Is it possible or would you expect that some of your Arizona customers could be only customers in Arizona? That some U.S. customers only want to buy wafers made in the U.S.?

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Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Sorry, your question is, will customers in Arizona only be U.S. customers?

Nicolas Baratte - *Macquarie Research - Analyst*

Is it possible some U.S. customers only want U.S.-made wafers?

C. C. Wei - *Taiwan Semiconductor Manufacturing Company Limited - CEO*

Well, let me answer that question. Arizona fab is for everybody, but majority is a U.S. customer, you are right.

Jeff Su - *Taiwan Semiconductor Manufacturing Company Limited - Director of IR*

Do you have another question? No? Okay. Well, if not, then this does conclude our Q&A session. Before we conclude today's conference, be advised that the replay of the conference will be accessible within 30 minutes from now, and the transcript will become available 24 hours from now, both of which are going to be available through our website, TSMC's website at www.tsmc.com.

So thank you again, everyone, for joining us today. We hope everyone continues to stay well, and we hope you will see -- join us again next quarter. Goodbye, and have a great day. Thanks.

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